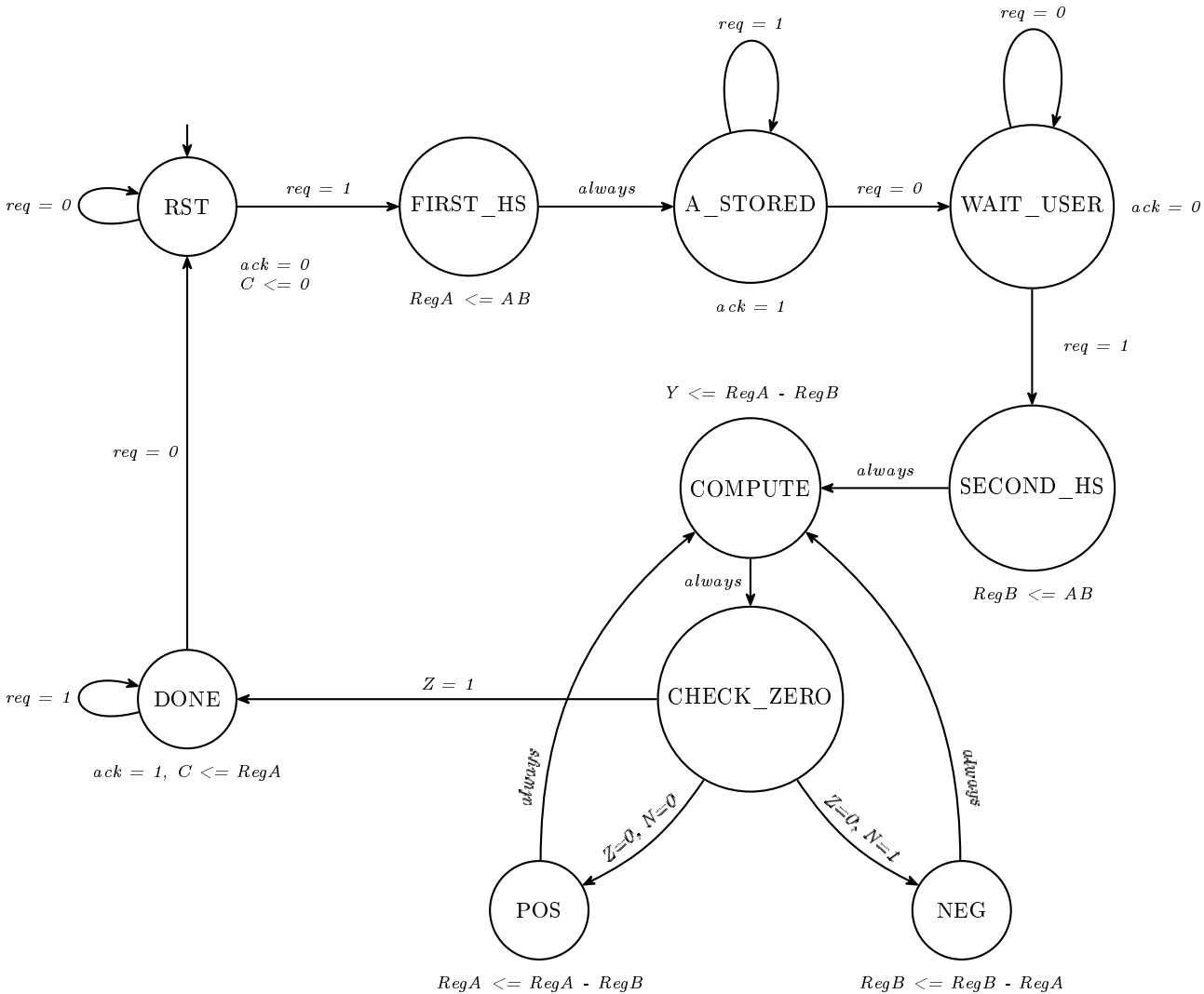
req = 0
RST
req = 1
ack = 0
C <= 0
FIRST_HS
RegA <= AB
always
req = 1
A_STORED
ack = 1
req = 0
WAIT_USER
ack = 0
req = 1
SECOND_HS
RegB <= AB
always
Y <= RegA - RegB
COMPUTE
always
CHECK_ZERO
Z = 1
req = 1
DONE
ack = 1, C <= RegA
req = 0
always
Z=0, N=0
POS
RegA <= RegA - RegB
Z=0, N=1
always
NEG
RegB <= RegB - RegA